

Comprehensive Linux Commands Cheat Sheet

Basic Navigation and File Operations

- `pwd` - Print working directory
- `ls` - List directory contents
- `cd <directory>` - Change directory
- `mkdir <directory>` - Create a directory
- `touch <file>` - Create an empty file
- `cp <source> <destination>` - Copy files or directories
- `mv <source> <destination>` - Move or rename files or directories

Viewing and Searching Files

- `head <file>` - Display the first 10 lines of a file
- `tail <file>` - Display the last 10 lines of a file
- `less <file>` - View file contents one screen at a time
- `grep <pattern> <file>` - Search for a pattern in a file
- `find <path> -name <filename>` - Search for files by name

File Permissions and Ownership

- `chmod <permissions> <file>` - Change file permissions (e.g., `chmod 755 file`)
- `chown <owner>:<group> <file>` - Change file owner and group
- `ls -l` - View detailed file permissions

Archiving and Compression

- `tar -cvf archive.tar <files>` - Create a tar archive
- `tar -xvf archive.tar` - Extract a tar archive
- `gzip <file>` - Compress a file
- `gunzip <file.gz>` - Decompress a file
- `zip <archive.zip> <files>` - Create a zip archive
- `unzip <archive.zip>` - Extract a zip archive

Process Management

- `ps` - Display running processes
- `top` - Interactive process viewer
- `htop` - Advanced interactive process viewer
- `kill <pid>` - Terminate a process by PID
- `killall <process_name>` - Terminate processes by name
- `bg` - Resume a process in the background
- `fg` - Resume a process in the foreground
- `jobs` - List background jobs

Networking Commands

- `ping <hostname>` - Test network connectivity
- `curl <url>` - Transfer data from a URL
- `wget <url>` - Download files from a URL
- `ssh <user>@<host>` - Connect to a remote server via SSH
- `scp <file> <user>@<host>:<destination>` - Copy files over SSH
- `netstat -tuln` - View listening ports

Disk Usage

- `df -h` - Display disk space usage
- `du -sh <directory>` - Display directory size
- `mount / umount` - Mount/Unmount file systems

System Monitoring and Troubleshooting

- dmesg - Display kernel messages
- journalctl - View system logs
- uptime - Show system uptime
- vmstat - Display system performance metrics
- iostat - Display I/O statistics
- free -h - Display memory usage
- lsof - List open files
- strace <command> - Trace system calls of a command

User Management

- `who` - Display logged-in users
- `whoami` - Show the current user
- `id` - Display user ID and group information
- `adduser <username>` - Add a new user
- `passwd <username>` - Change user password
- `usermod -aG <group> <username>` - Add user to a group

Firewall and Security

- `ufw status` - Check firewall status
- `ufw allow <port>` - Allow traffic on a port
- `ufw deny <port>` - Deny traffic on a port
- `iptables -L` - List firewall rules
- `fail2ban-client status` - Check Fail2Ban status

Advanced Networking and Security

- `traceroute <host>` - Trace the route to a host
- `nslookup <domain>` - Query DNS information
- `iptables -L` - List firewall rules
- `tcpdump` - Network packet analyzer
- `openssl` - Create and manage certificates
- `nmap` - Network scanning and security auditing

Advanced File Operations and Scripting

- `rsync -av <source> <destination>` - Sync files/directories efficiently
- `awk / sed` - Advanced text processing
- `bash` scripting - Automate repetitive tasks

Monitoring and Diagnostics

- sar - System Activity Report
- iotop - Show I/O usage by processes
- perf - Performance analysis tool

Package Management

- Debian-based systems:
- `apt update` - Update package lists
- `apt install <package>` - Install a package
- `apt remove <package>` - Remove a package
- RHEL-based systems:
- `yum install <package>` - Install a package
- `yum remove <package>` - Remove a package
- `dnf update` - Update packages

Performance Tuning

- `nice -n <priority> <command>` - Set process priority
- `renice <priority> <pid>` - Change priority of a running process
- `sysctl -a` - Display kernel parameters
- `ulimit -a` - Display user limits

Docker and Containers

- `docker ps` - List running containers
- `docker run <image>` - Run a container
- `kubectl get pods` - Interact with Kubernetes clusters